







MiniPoly: Automatic Extraction of Efficient JavaScript Polyfill from Language Specification

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2026.10.13





(* equally contributed)

Built-in APIs in JavaScript

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const a = new Set([1, 2, 3]);
const b = new Set([1, 2]);
a.isSupersetOf(b);   // true
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≥ 122



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ES2025

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≤ 126





≤ 2025.09





≥ 122





≤ 16.7





≤ 21



Built-in APIs in JavaScript

ES2025

Native Code
(e.g., C++, Java)

```
const a = new Set([1, 2, 3]);
const b = new Set([1, 2]);
a.isSupersetOf(b);   // true
```

```
Welcome to Node.js v24.14.1.
Type ".help" for more information.
> Set.prototype.isSupersetOf.toString()
'function isSupersetOf() { [native code] }'
>
```



≤ 126



≤ 2025.09



≥ 122



≤ 16.7



≤ 21

Polyfill

- **Polyfill** is a **JavaScript** implementation of built-in APIs
- **core-js** is used by **41% of websites** and has **≥ 46M downloads per week**

```
module.exports = function isSupersetOf(other) {
  var O = aSet(this);
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  return iterateSimple(iterator, function (e) {
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core-js

Polyfill

```
import 'core-js/actual/set/is-superset-of';
const a = new Set([1, 2, 3]);
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core-js



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ECMA-262 - JavaScript Language Specification



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Syntax

AdditiveExpression`[Yield, Await]` :

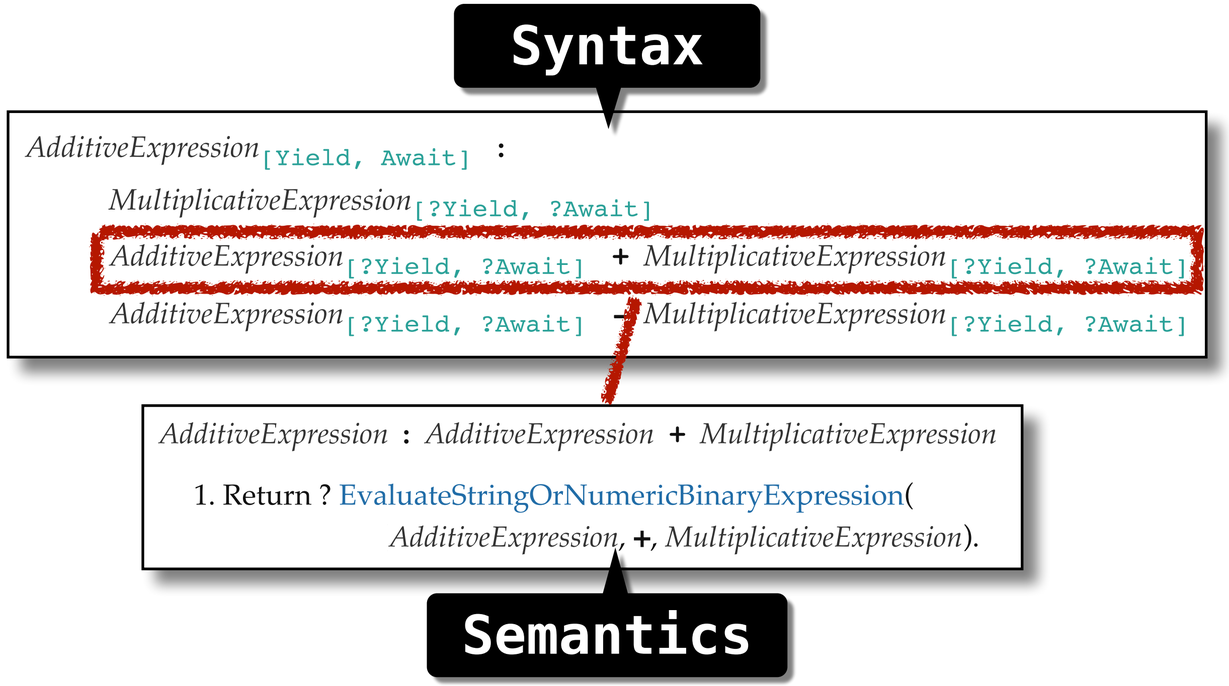
MultiplicativeExpression`[?Yield, ?Await]`

AdditiveExpression`[?Yield, ?Await]` + *MultiplicativeExpression*`[?Yield, ?Await]`

AdditiveExpression`[?Yield, ?Await]` - *MultiplicativeExpression*`[?Yield, ?Await]`

ECMA-262 - JavaScript Language Specification





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TC 39

tc39/**ecma262**

Status, process, and documents for ECMA-262

TC 39

Syntax

AdditiveExpression`[Yield, Await]` :

MultiplicativeExpression`[?Yield, ?Await]`

AdditiveExpression`[?Yield, ?Await]` + *MultiplicativeExpression*`[?Yield, ?Await]`

AdditiveExpression`[?Yield, ?Await]` - *MultiplicativeExpression*`[?Yield, ?Await]`

AdditiveExpression : *AdditiveExpression* + *MultiplicativeExpression*

1. Return ? EvaluateStringOrNumericBinaryExpression(*AdditiveExpression*, +, *MultiplicativeExpression*).

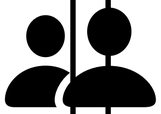
Semantics

Current Approach - **Manual Translation**

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Manual

TC 39

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core-js

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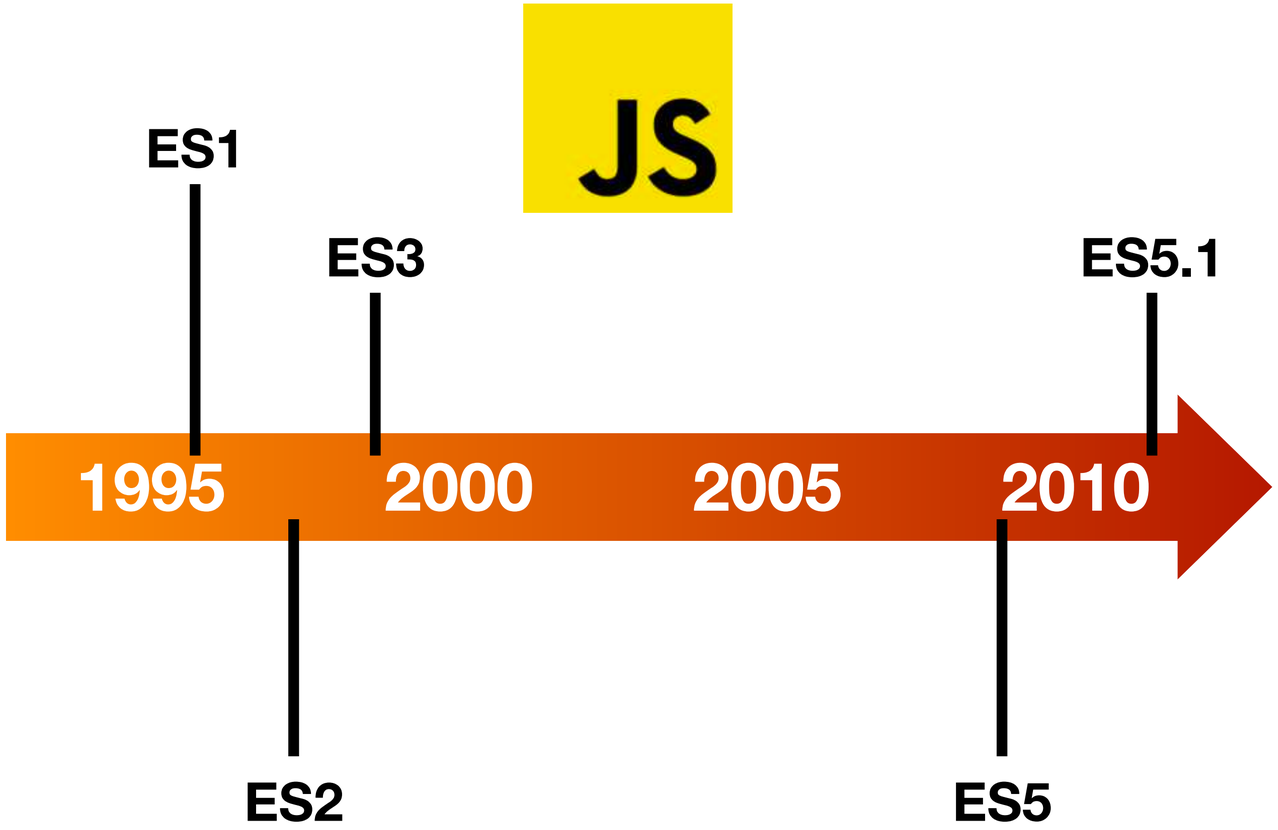
TC 39

Manual

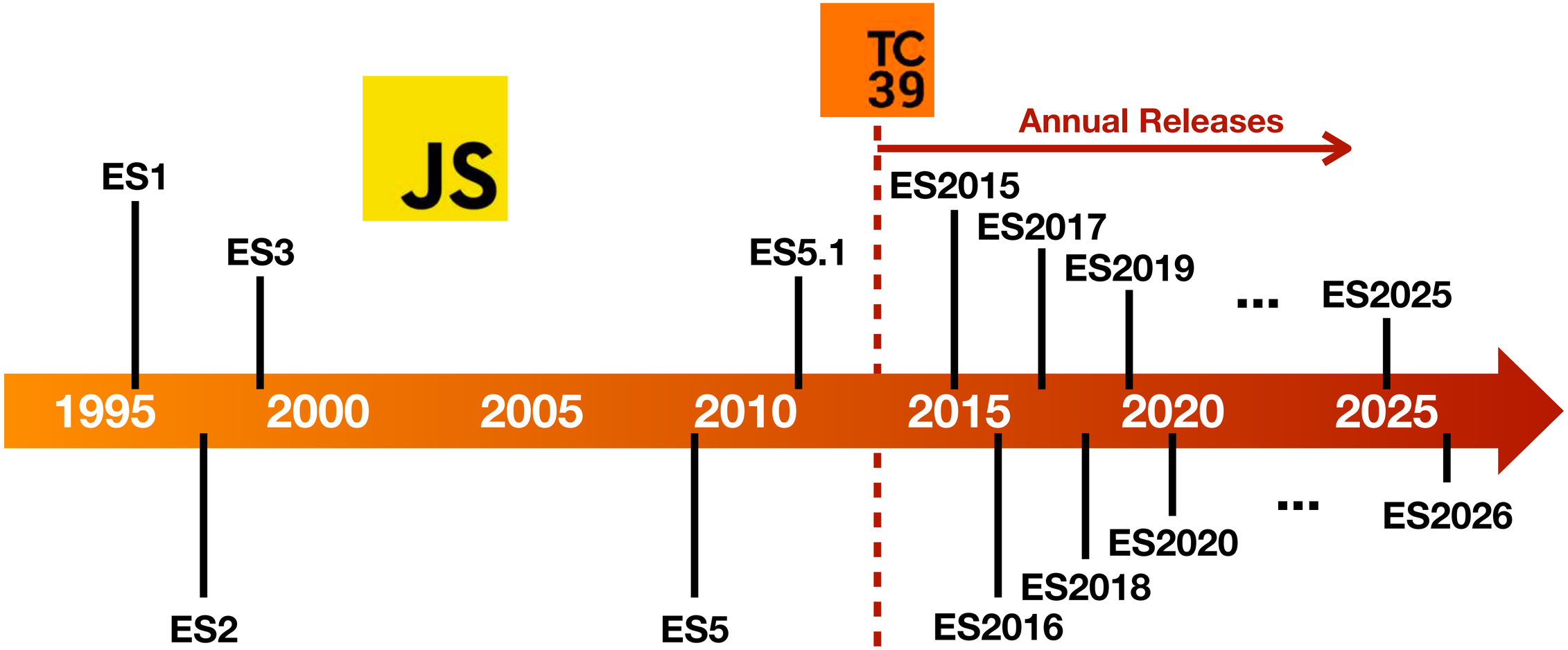
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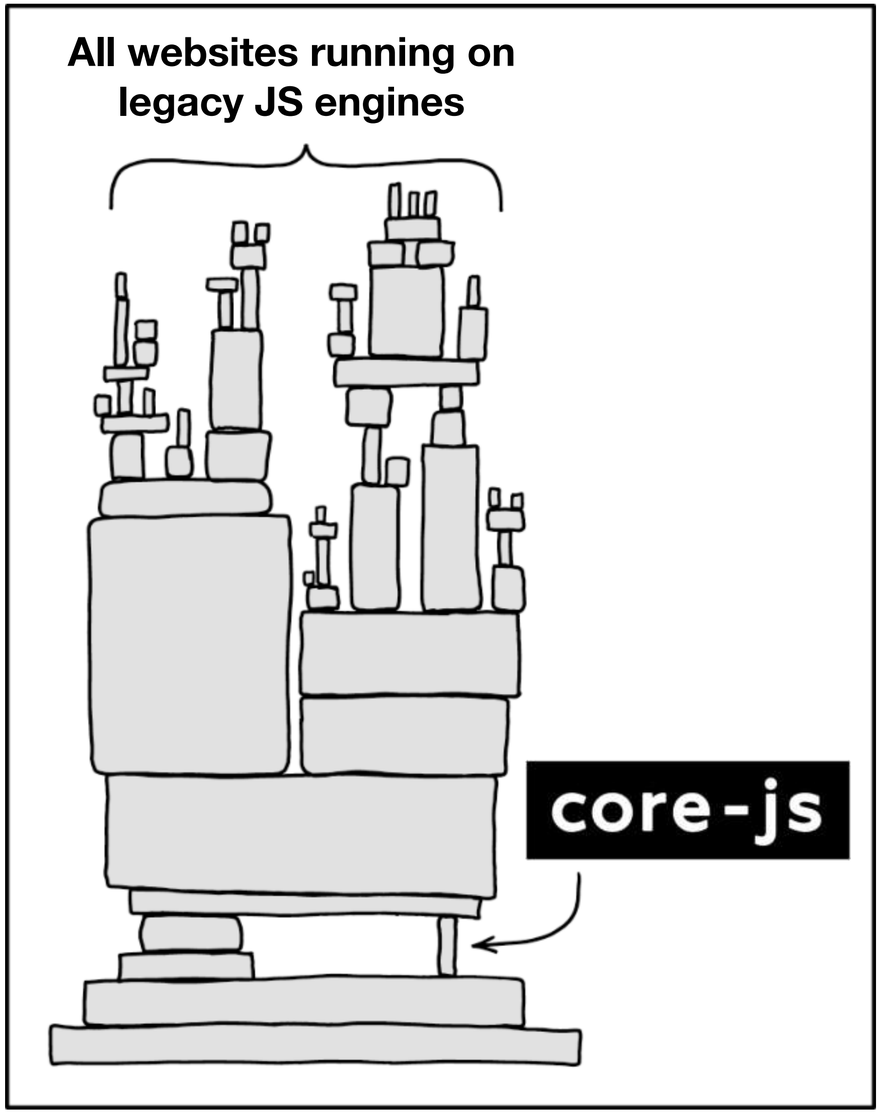
Problem - **Fast Evolving** JavaScript



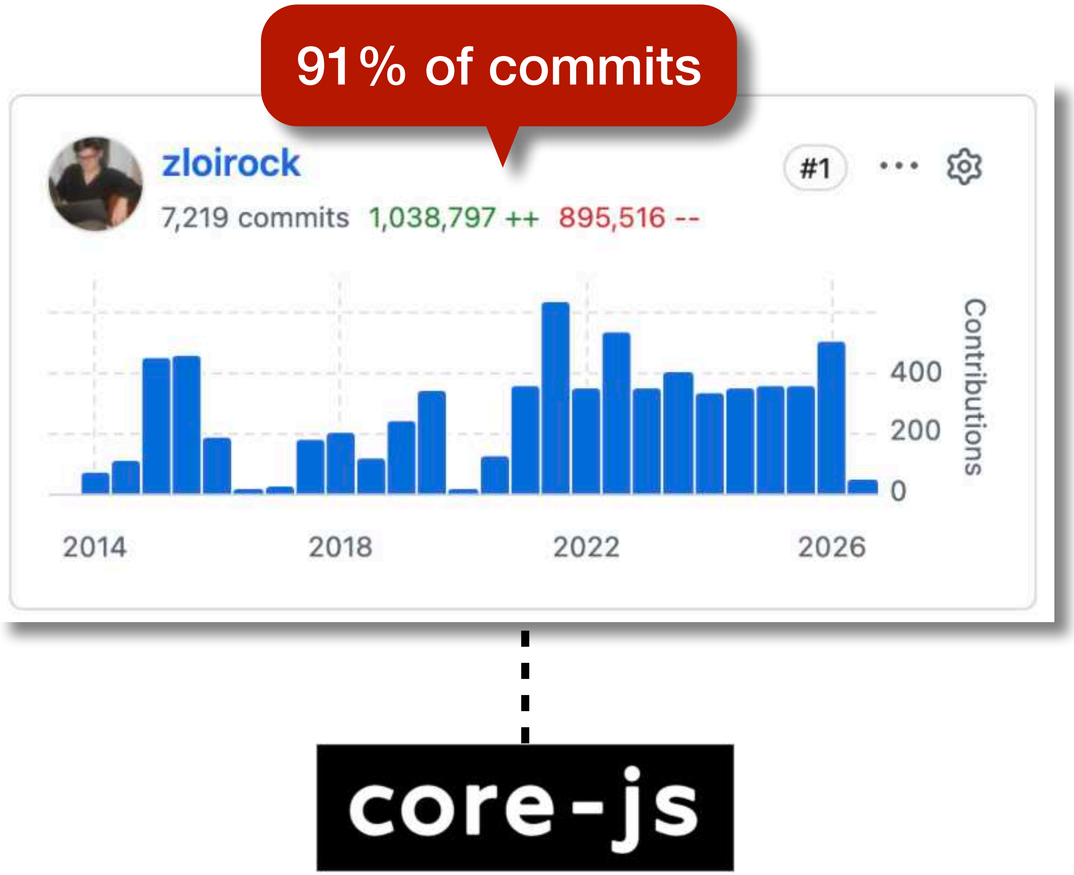
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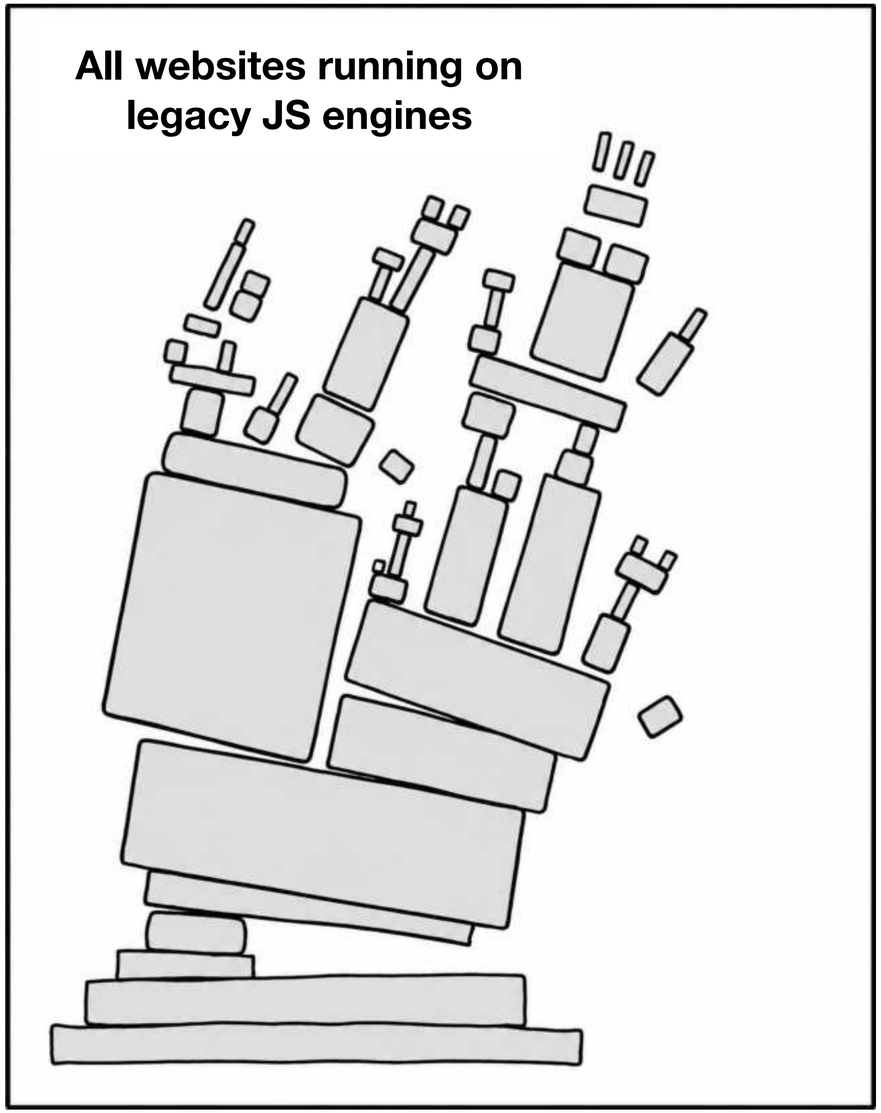
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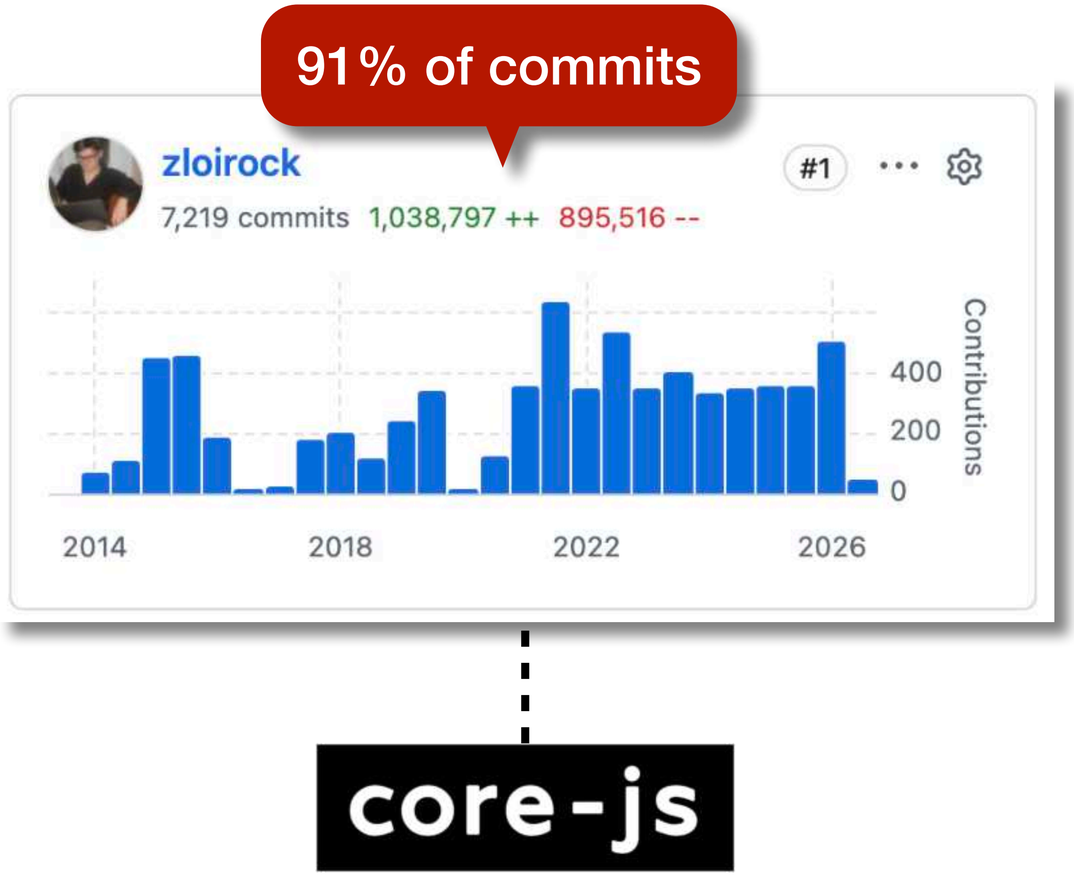
https://xkcd.com/2347/ (Original, modified with AI)



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Our Approach - **Automatic Extraction** from ECMA-262

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Extract

TC 39

Polyfill in **JavaScript**

Challenges - **Correctness** and **Efficiency**

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Extract

Polyfill in **JavaScript**

TC 39

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1) How to **correctly** handle *internal repr. of control flows*?

2) How to extract *algorithmically* **efficient** polyfill?

Extract

Polyfill in **JavaScript**

TC 39

Challenge 1 - **Correctness** (Internal Repr. of Control Flow)

A computation **X**

- is **normally** terminated with **a value** — **if X** is a **normal completion**
- **throws** an **exception** — **if X** is a **throw completion**

IteratorClose (*iteratorRecord*, *completion*)
...
3. Let *innerResult* be **GetMethod**(...).
4. If *innerResult* is a **normal completion**, then
   ...
   b. If *return* is undefined, return ? *completion*.
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5. If *completion* is a **throw completion**, return ? *completion*.
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TC 39

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```
return completion
   OR
throw completion
```

TC 39

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`return completion`
OR
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`throw completion`

`throw innerResult`

TC 39

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TC 39

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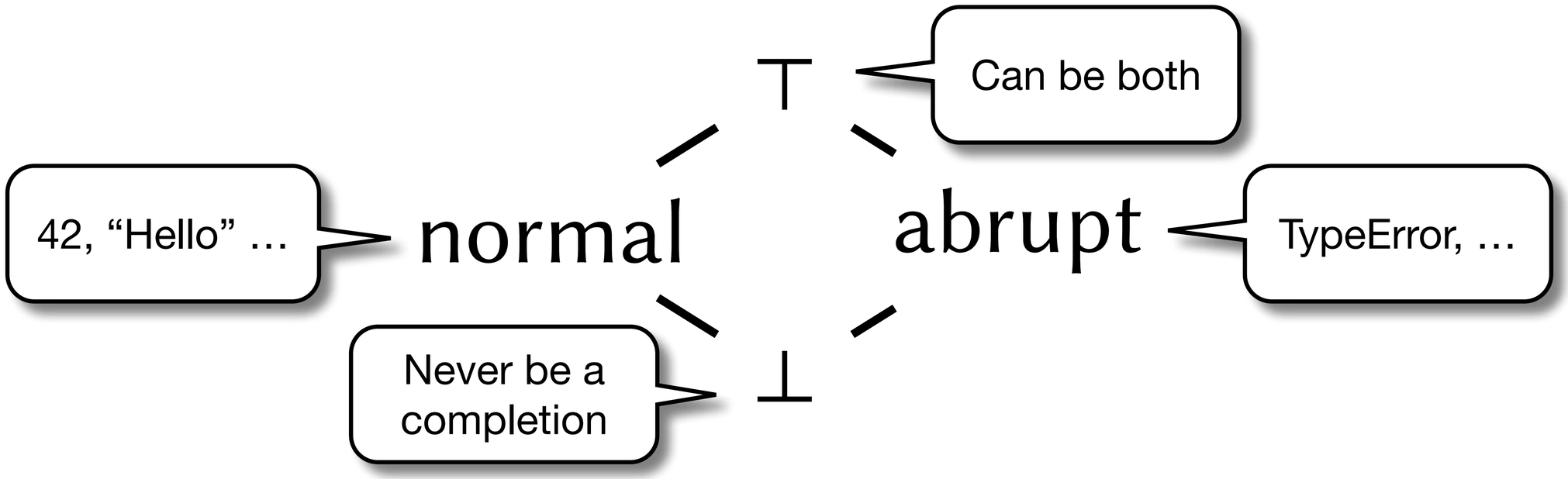
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`return completion`

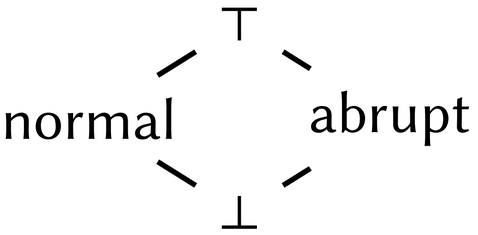
Solution 1 - **Completion Type Analysis**

IteratorClose (*iteratorRecord*, *completion*)
...
3. Let *innerResult* be **GetMethod**(...).

TC 39



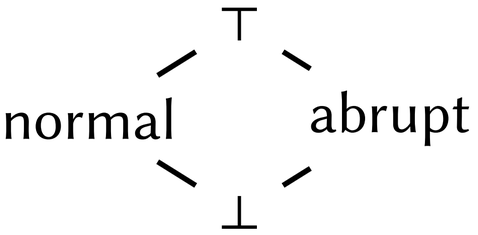
Solution 1 - **Completion Type Analysis**



IteratorClose (*iteratorRecord*, *completion*)
...
3. Let *innerResult* be **GetMethod**(...).

TC 39

Solution 1 - **Completion Type Analysis**



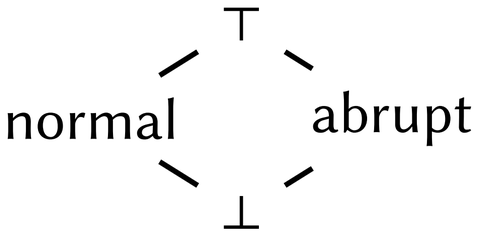
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⊤

⊤

TC 39

Solution 1 - **Completion Type Analysis**



IteratorClose (*iteratorRecord*, *completion*)
...
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⊤

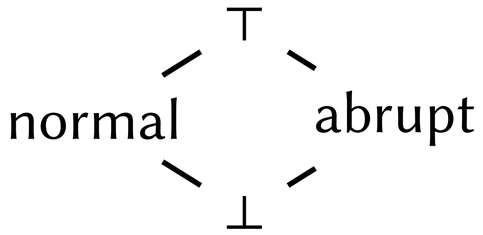
⊤

TC 39

```
function IteratorClose(iteratorRecord, completion, completion_flag) {
  ...
  try {
    var innerResult = GetMethod(...);
    var innerResult_flag = true;
  } catch (e) {
    var innerResult = e;
    var innerResult_flag = false;
  }
```

JS

Solution 1 - **Completion Type Analysis**



IteratorClose (*iteratorRecord*, *completion*)
...
3. Let *innerResult* be **GetMethod**(...).

completion : ⊤
innerResult : ⊤

⊤

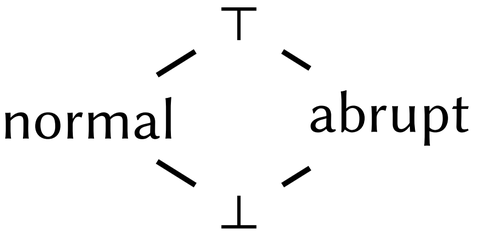
⊤

TC 39

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JS

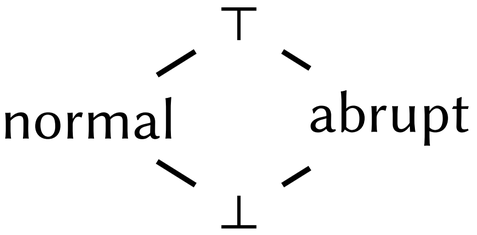
Solution 1 - **Completion Type Analysis**



4. If *innerResult* is a **normal completion**, then
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   b. If *return* is undefined, return ? *completion*
   c. Set *innerResult* to **Call**(...).

TC 39

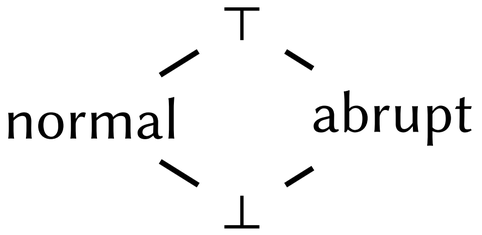
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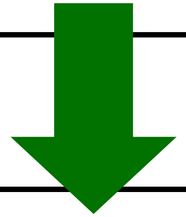
TC 39

Solution 1 - **Completion Type Analysis**



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TC 39



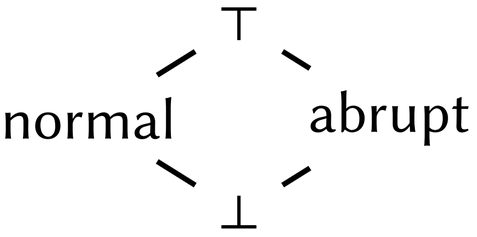
```
if (innerResult_flag) {
  ...



}
```

JS

Solution 1 - **Completion Type Analysis**



4. If *innerResult* is a **normal completion**, then
   ...
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completion : ⊤
innerResult : normal

TC 39

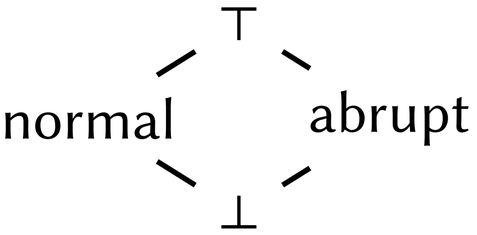
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if (innerResult_flag) {
  ...



}
```

JS

Solution 1 - **Completion Type Analysis**



4. If *innerResult* is a **normal completion**, then
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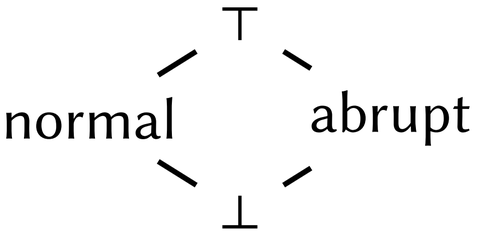
TC 39

```
if (innerResult_flag) {
  ...
  if ( return === undefined) {
    if (completion_flag) return completion;
    else                 throw  completion;
  }

}
```

JS

Solution 1 - **Completion Type Analysis**



4. If *innerResult* is a **normal completion**, then
   ...
   b. If *return* is undefined, return ? *completion*
   c. Set *innerResult* to **Call**(...)

completion : ⊤
innerResult : normal

⊤

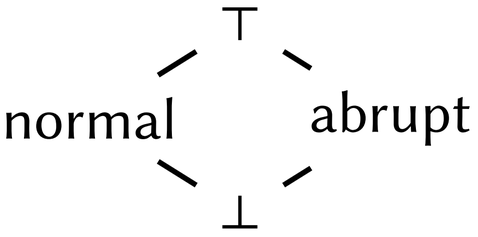
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JS

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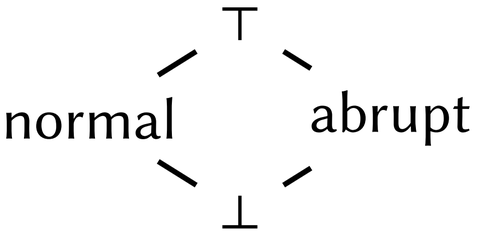
⊤

TC 39

```
if (innerResult_flag) {
  ...
  if ( return === undefined) {
    if (completion_flag) return completion;
    else                 throw  completion;
  }
  try { innerResult = Call(...); innerResult_flag = true; } catch ...
}
```

JS

Solution 1 - **Completion Type Analysis**



4. If *innerResult* is a **normal completion**, then
   ...
   b. If *return* is undefined, return ? *completion*
   c. Set *innerResult* to **Call**(...)

completion : ⊤
innerResult : normal

innerResult : ⊤

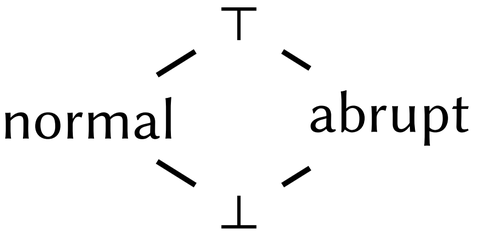
⊤

TC 39

```
if (innerResult_flag) {
  ...
  if ( return === undefined) {
    if (completion_flag) return completion;
    else                 throw  completion;
  }
  try { innerResult = Call(...); innerResult_flag = true; } catch ...
}
```

JS

Solution 1 - **Completion Type Analysis**



5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.

...

8. Return ? *completion*.

TC 39

Solution 1 - **Completion Type Analysis**



5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.
...
8. Return ? *completion*.

TC 39

completion : abrupt

innerResult : abrupt

Solution 1 - **Completion Type Analysis**



5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.
...
8. Return ? *completion*.

TC39

completion : abrupt

innerResult : abrupt

```
if (!completion_flag)  throw completion;

if (!innerResult_flag) throw innerResult;
```

JS

Solution 1 - **Completion Type Analysis**



5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.
...
8. Return ? *completion*.

completion : abrupt

innerResult : abrupt

TC39

completion : normal

```
if (!completion_flag)  throw completion;

if (!innerResult_flag) throw innerResult;
```

JS

Solution 1 - **Completion Type Analysis**



5. If *completion* is a **throw completion**, return ? *completion*.
6. If *innerResult* is a **throw completion**, return ? *innerResult*.
...
8. Return ? *completion*.

completion : abrupt

innerResult : abrupt

TC39

completion : normal

```
if (!completion_flag)  throw completion;

if (!innerResult_flag) throw innerResult;

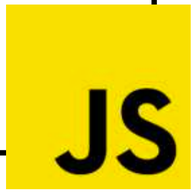
...

return completion;
```

JS

Solution 1 - **Completion Type Analysis**

```
function IteratorClose(iteratorRecord, completion, completion_flag) {
  ...
  try { var innerResult = GetMethod(...); var innerResult_flag = true; }
  catch (e) { var innerResult = e; var innerResult_flag = false; }
  if (innerResult_flag) {
    ...
    if (_return === undefined) {
      if (completion_flag) return completion;
      else                 throw  completion;
    }
    try { innerResult = Call(...); innerResult_flag = true; } catch ...
  }
  if (!completion_flag)  throw completion;
  if (!innerResult_flag) throw innerResult;
  ...
  return completion;
}
```



Challenge 2 - **Efficiency**

TC 39

Set.prototype.add (*value*)
...
4. For each element *e* of *S*.[[SetData]], do
  a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then
  ...

Challenge 2 - **Efficiency**

Set.prototype.add (*value*)

Linear search

TC 39

...

4. For each element *e* of *S*.[[SetData]], do
   a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then
   ...

Challenge 2 - **Efficiency**

$$\mathcal{O}(n)$$

Set.prototype.add (*value*)

Linear search

TC 39

...

4. For each element *e* of *S*.[[SetData]], do
   a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then
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Challenge 2 - **Efficiency**

$\mathcal{O}(n)$

Set.prototype.add (*value*)

Linear search

...

4. For each element *e* of *S*.[[SetData]], do

   a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then

   ...

TC 39

vs.

$\mathcal{O}(1)$

core-js

Set.prototype.add

Challenge 2 - **Efficiency**

$\mathcal{O}(n)$ **vs.** $\mathcal{O}(1)$

TC 39

Linear search

Set.prototype.add (*value*)
...
4. For each element *e* of *S*.[[SetData]], do
  a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then

Set.prototype.delete (*value*)
...
4. For each element *e* of *S*.[[SetData]], do
  a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then

Set.prototype.has (*value*)
...
4. For each element *e* of *S*.[[SetData]], do
  a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then
  ...

core-js

Set.prototype.add

Set.prototype.delete

Set.prototype.has

Solution 2 - **Rewrite Rules**

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not ~empty~ and SameValue({{ elem }},
          {{ value : var }}) is *true*, then
      1. {{ body : step }}
```

(* simplified)

Set.prototype.add (*value*)

...

4. For each element *e* of *S*.[[SetData]], do
   a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then
   ...

TC 39

Solution 2 - **Rewrite Rules**

```
pattern: |
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    1. If {{ elem }} is not ~empty~ and SameValue({{ elem }},
          {{ value : var }}) is *true*, then
      1. {{ body : step }}
where:
  - isSetData(base)
  - { isSetData: "_.[[SetData]].copy*" }
```

(* simplified)

Set.prototype.add (*value*)

...

4. For each element *e* of *S*.[[SetData]], do
   a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then
   ...

TC 39

Solution 2 - **Rewrite Rules**

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pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not ~empty~ and SameValue({{ elem }},
          {{ value : var }}) is *true*, then
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where:
  - isSetData(base)
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(* simplified)

Set.prototype.add (*value*)

...

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   ...

TC 39

Solution 2 - **Rewrite Rules**

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not ~empty~ and SameValue({{ elem }},
          {{ value : var }}) is *true*, then
      1. {{ body : step }}
where:
  - isSetData(base)
  - { isSetData: " _.[[SetData]].copy* " }
replace: |
  1. If SET_DATA_HAS({{ base }}, {{ value }}) is *true*, then
    1. {{ body }}
```

(* simplified)

Set.prototype.add (*value*)

...

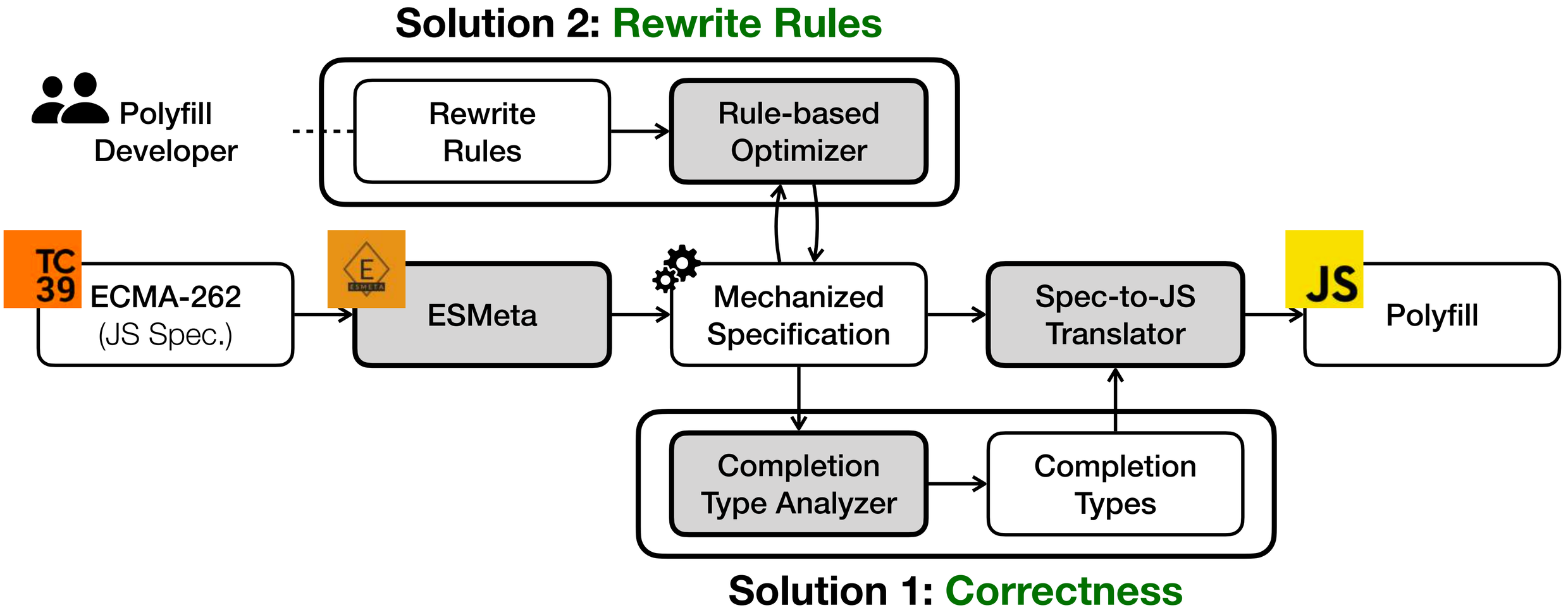
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   a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then
   ...

TC 39

```
function add(value) {
  ...
  if (SET_DATA_HAS(S.SetData, value)
      === true) {
    ...
  }
}
```

JS

MiniPoly - Overall Structure



Evaluation

- Target **89 built-in APIs** in **ES2025**
- **Baselines** - `core-js` and `es-shims` (final release versions of 2025)
- Research Questions

  RQ1) Correctness

  RQ2) Reduced Manual Effort

  RQ3) Speed

  RQ4) Forward Compatibility

  RQ5) Real-World Usability

RQ1) Correctness

- Tested by **Test262** - official conformance tests
- **(# of Fail) core-js** - 344 / **es-shims** - 618 (of 6,785 supported) / **MiniPoly** - 16

			core-js		es-shims†		MiniPoly		MiniPoly-opt	
Category	**# Built-ins**	**Tests**	**Pass (%)**	**Time**	**Pass (%)**	**Time**	**Pass (%)**	**Time**	**Pass (%)**	**Time**
Array	24	4,713	4,617 (98.0)	26.4s	4,355 (94.9)	48.5s	**4,713 (100.0)**	25.4s	**4,713 (100.0)**	25.6s
String	12	638	616 (96.6)	6.2s	602 (94.4)	6.6s	**638 (100.0)**	6.0s	**638 (100.0)**	6.0s
Set	18	748	710 (94.9)	8.5s	682 (91.7)	9.4s	**746 (99.7)**	8.2s	**746 (99.7)**	8.4s
Map	12	322	298 (92.5)	4.8s	256 (79.5)	5.0s	**320 (99.4)**	4.7s	**320 (99.4)**	4.6s
WeakMap	5	198	168 (84.8)	2.3s	–	–	**196 (99.0)**	2.2s	**196 (99.0)**	2.3s
WeakSet	5	166	142 (85.5)	2.2s	–	–	**164 (98.8)**	2.1s	**164 (98.8)**	2.1s
Promise	13	1,264	1,154 (91.3)	9.8s	272 (55.5)	19.8s	**1,256 (99.4)**	9.6s	**1,256 (99.4)**	9.6s
Total	**89**	**8,049**	**7,705 (95.7)**	**60.1s**	**6,167 (90.9)**	**89.3s**	**8,033 (99.8)**	**58.1s**	**8,033 (99.8)**	**58.4s**

† es-shims evaluated on supported subset, and (–) indicates unsupported built-ins.

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WeakSet	5	166	142 (85.5)	2.2s	–	–	**164 (98.8)**	2.1s	**164 (98.8)**	2.1s
Promise	13	1,264	1,154 (91.3)	9.8s	272 (55.5)	19.8s	**1,256 (99.4)**	9.6s	**1,256 (99.4)**	9.6s
Total	**89**	**8,049**	**7,705 (95.7)**	**60.1s**	**6,167 (90.9)**	**89.3s**	**8,033 (99.8)**	**58.1s**	**8,033 (99.8)**	**58.4s**

† es-shims evaluated on supported subset, and (–) indicates unsupported built-ins.

RQ1) Correctness

Inherent limitations shared by all polyfills

- Tested by **Test262** - official conformance tests
- **(# of Fail) core-js - 344 / es-shims - 618** (of 6,785 supported) **/ MiniPoly - 16**

			core-js		es-shims†		MiniPoly		MiniPoly-opt	
Category	**# Built-ins**	**Tests**	**Pass (%)**	**Time**	**Pass (%)**	**Time**	**Pass (%)**	**Time**	**Pass (%)**	**Time**
Array	24	4,713	4,617 (98.0)	26.4s	4,355 (94.9)	48.5s	**4,713 (100.0)**	25.4s	**4,713 (100.0)**	25.6s
String	12	638	616 (96.6)	6.2s	602 (94.4)	6.6s	**638 (100.0)**	6.0s	**638 (100.0)**	6.0s
Set	18	748	710 (94.9)	8.5s	682 (91.7)	9.4s	**746 (99.7)**	8.2s	**746 (99.7)**	8.4s
Map	12	322	298 (92.5)	4.8s	256 (79.5)	5.0s	**320 (99.4)**	4.7s	**320 (99.4)**	4.6s
WeakMap	5	198	168 (84.8)	2.3s	–	–	**196 (99.0)**	2.2s	**196 (99.0)**	2.3s
WeakSet	5	166	142 (85.5)	2.2s	–	–	**164 (98.8)**	2.1s	**164 (98.8)**	2.1s
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† es-shims evaluated on supported subset, and (–) indicates unsupported built-ins.

RQ2) Reduced Manual Effort

		Baseline		MiniPoly		MiniPoly-opt		
Object	**#**	**core-js**	**es-shims**	**Auto**	**Man.**	**Auto**	**Man.**	**Rules**
Array	24	2,298	8,978	1,332	317			
String	12	1,281	6,885	418	130			
Set	18	2,313	5,880	864	190	861	283	114 (11)
WeakSet	5	1,882	–	387	123	379	214	
Map	12	1,986	3,062	756	217	747	327	81 (6)
WeakMap	5	2,034	–	446	115	436	231	
Promise	13	2,518	8,031	1,511	203			
Total†	**89**	**14,312**	**32,836**	**5,714**	**1,295**	**5,684**	**1,705**	**195 (17)**

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3.94%

9.05%

RQ2) Reduced Manual Effort

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17 Manual Rules

RQ2) Reduced Manual Effort

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17 Manual Rules

13.28%

RQ2) Reduced Manual Effort

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Total†	**89**	**14,312**	**32,836**	**5,714**	**1,295**	**5,684**	**1,705**	**195 (17)**

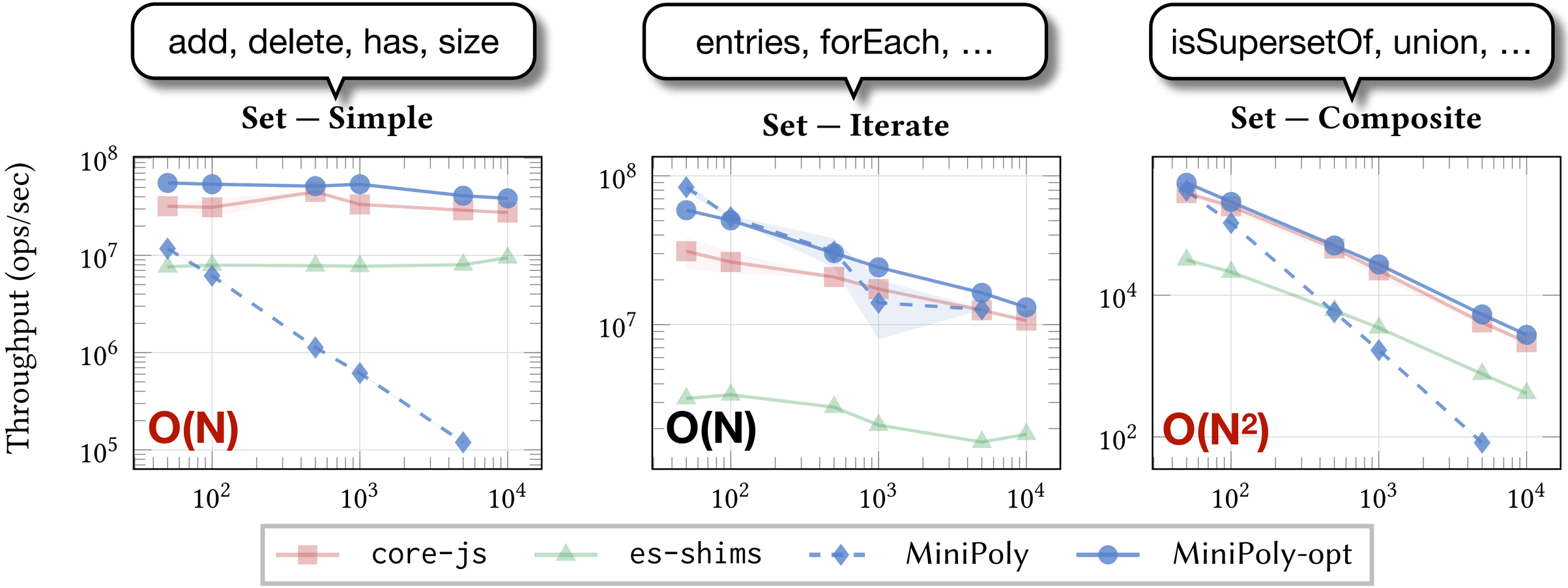
5.79%

13.28%

17 Manual Rules

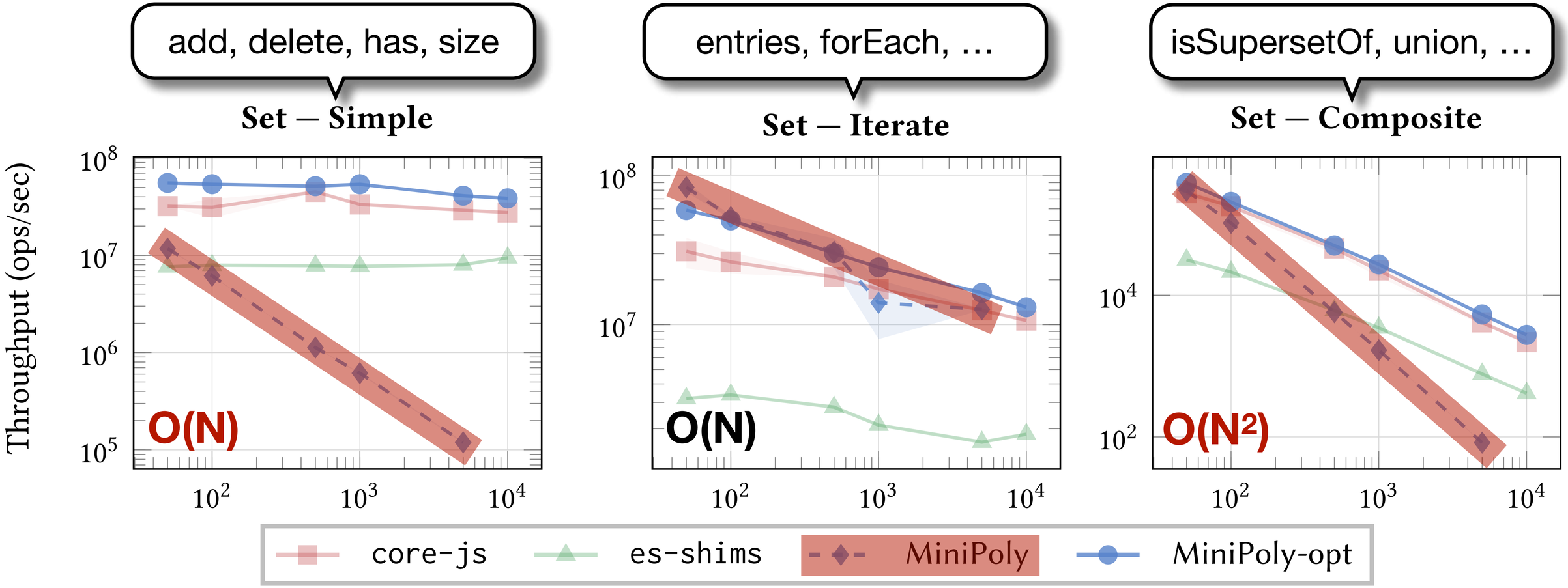
RQ3) Speed

- We measured **throughput** (*y-axis* = ops/sec) across **sizes** (*x-axis* = N)



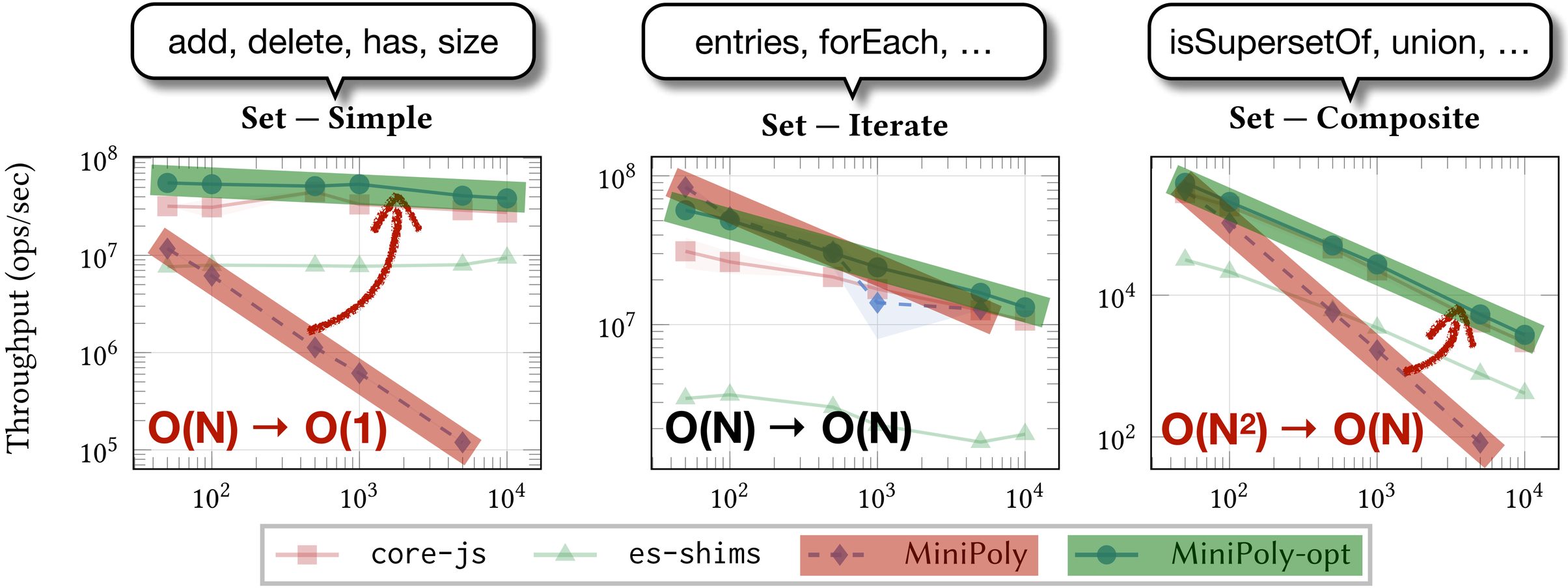
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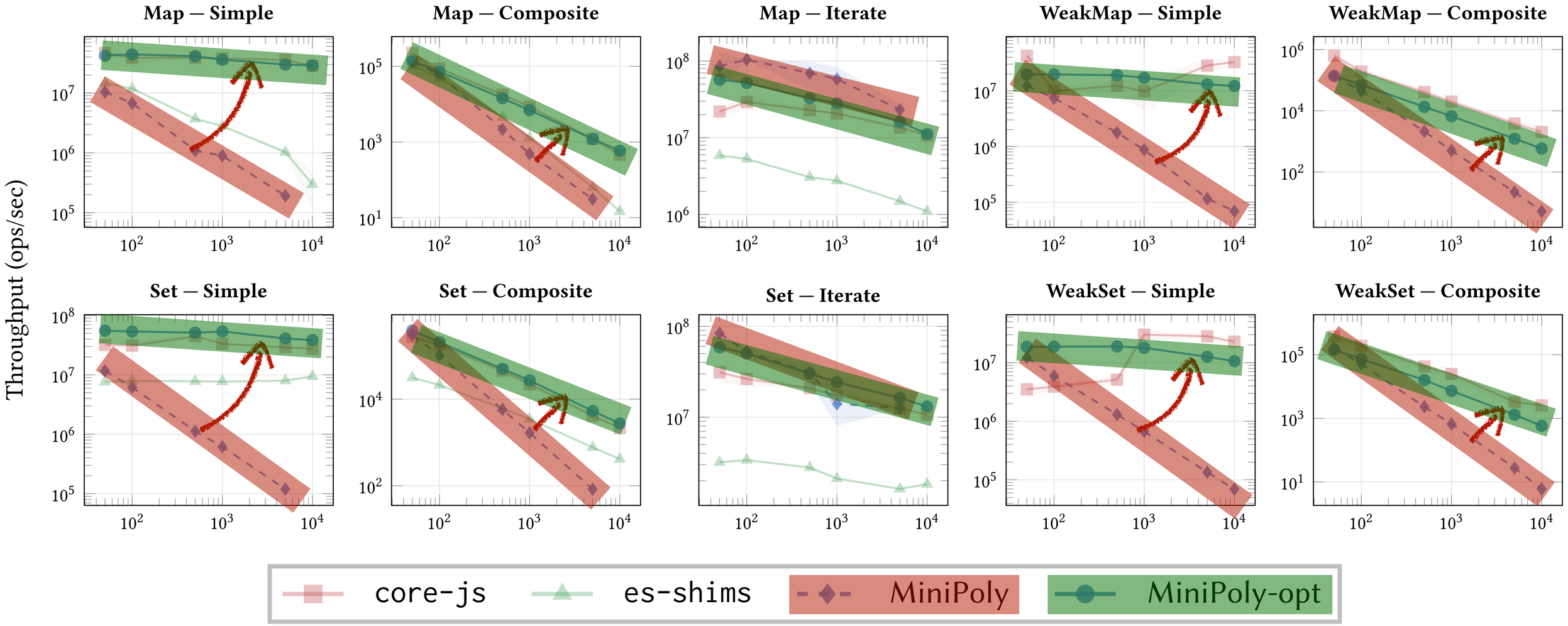
RQ3) Speed

- We measured **throughput** (*y-axis* = ops/sec) across **sizes** (*x-axis* = N)



RQ3) Speed

Competitive with baselines in most cases



RQ4) Forward Compatibility

- Supports Built-in APIs in **ES2026** and **Stage 2.7**

  with **100% correctness** and **comparable speed** with manual baselines

Proposal	Method	Tests	Correctness (Pass %)				Throughput (ops/sec) — N = 5,000			
			core-js	es-shims	MiniPoly	MiniPoly-opt	core-js	es-shims	MiniPoly	MiniPoly-opt
Upsert (ES2026)	Map.prototype									
	.getOrInsert	28	26 (92.9)	24 (85.7)	**28 (100.0)**	**28 (100.0)**	13.7M / 2.0M	61.2K	107K	8.2M / 2.5M
	.getOrInsertComputed	37	35 (94.6)	33 (89.2)	**37 (100.0)**	**37 (100.0)**	11.7M / 1.8M	40.8K	77.7K	6.5M / 1.9M
	WeakMap.prototype									
	.getOrInsert	34	26 (76.5)	—	**34 (100.0)**	**34 (100.0)**	18.4M / 3.1M	—	104K	8.2M / 2.3M
	.getOrInsertComputed	43	32 (74.4)	—	**43 (100.0)**	**43 (100.0)**	5.7M / 1.2M	—	73.1K	5.0M / 1.8M
Await Dictionary (Stage 2.7)	Promise									
	.allKeyed	12	—	—	**12 (100.0)**	**12 (100.0)**	—	—	—	—
	.allSettledKeyed	12	—	—	**12 (100.0)**	**12 (100.0)**	—	—	—	—
Total		**166**	**119 (83.8)**	**57 (87.7)**	**166 (100.0)**	**166 (100.0)**	11.3M / 1.9M	50.0K	89.3K	6.8M / 2.1M

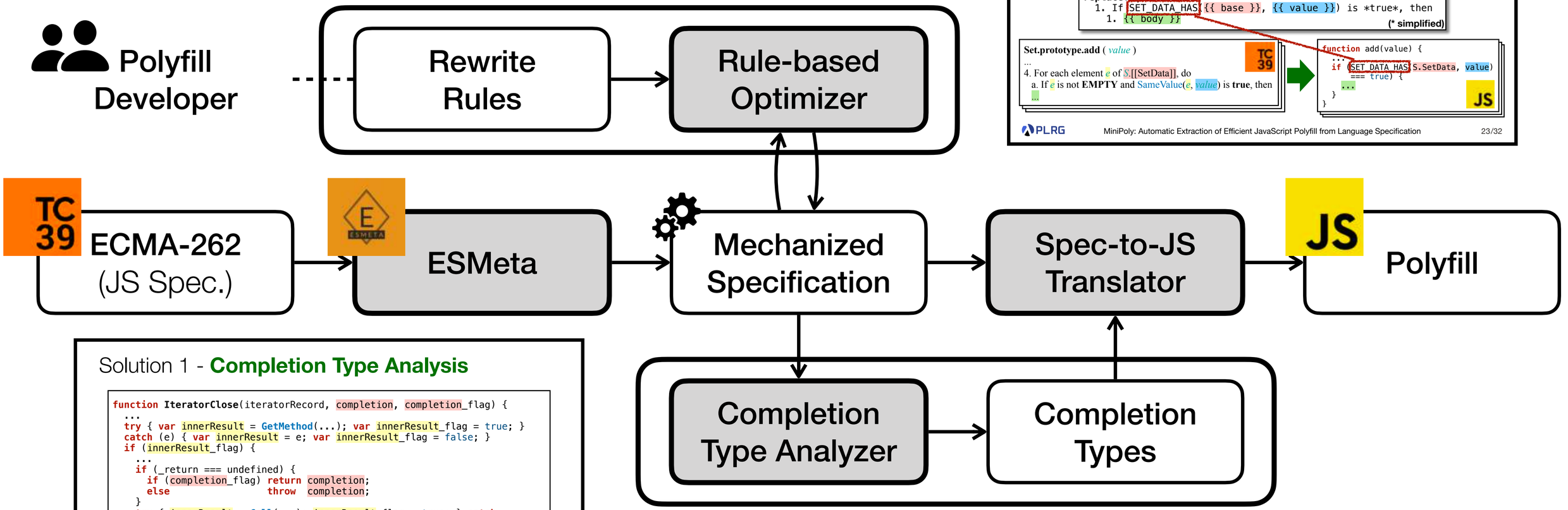
-23 -8

KOREA UNIVERSITY

PLRG

Challenge 2: Efficiency

Polyfill Developer



Solution 2 - Rewrite Rules

```
pattern: |
  1. For each element {{ elem : var }} of {{ base : ref }}, do
    1. If {{ elem }} is not ~empty~ and SameValue({{ elem }},
          {{ value : var }}) is *true*, then
      1. {{ body : step }}
where:
  - isSetData(base)
  - { isSetData: " _.[[SetData]] " }
replace: |
  1. If SET_DATA_HAS({{ base }}, {{ value }}) is *true*, then
    1. {{ body }}
```

(* simplified)

Set.prototype.add (*value*)
...
4. For each element *e* of *S*.[[SetData]], do
  a. If *e* is not **EMPTY** and SameValue(*e*, *value*) is **true**, then
  ...

```
function add(value) {
  ...
  if (SET_DATA_HAS(S.SetData, value)
      === true) {
    ...
  }
}
```


Solution 1 - Completion Type Analysis

```
function IteratorClose(iteratorRecord, completion, completion_flag) {
  ...
  try { var innerResult = GetMethod(...); var innerResult_flag = true; }
  catch (e) { var innerResult = e; var innerResult_flag = false; }
  if (innerResult_flag) {
    ...
    if (_return === undefined) {
      if (completion_flag) return completion;
      else                 throw  completion;
    }
    try { innerResult = Call(...); innerResult_flag = true; } catch ...
  }
  if (!completion_flag)  throw completion;
  if (!innerResult_flag) throw innerResult;
  ...
  return completion;
}
```


Challenge 1: Correctness